University of Barishal
Department of Computer Science and Engineering

Course Code: CSE-4101
Course Title: Artificial Intelligence
4th Year 1st Semester
Admission Session: 2016-2017

Time: 03 Hours **Marks:** 60

Answer any **FIVE** questions out of the followings. All parts of each question must be answered consecutively. Right side of the question shows the maximum marks.

1. a) Define Artificial Intelligence (AI). Describe Turing test approach. [3]

   b) Define Agent and Rational Agent through real time example [3]

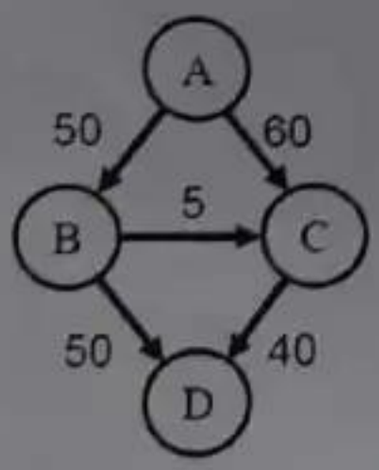
   c) Distinguish between the following properties of a task environment: [3]
      i) Static or dynamic
      ii) Discrete or continuous
      iii) Single or multi-agent

   d) Briefly explain the PEAS description of the task environment for **an automated taxi, Part Picking Robot and Refinery Controller.** [3]

2. a) Write short notes on agent type: [3]
      i) Simple reflex agents ii) Model-based reflex agents;

   b) Explain difference between uninformed and informed search. List two examples of each type of algorithm. Proof that Uniform-cost search is a special case of A* search. [4]

   c) The state space description for a problem is shown below, with **A** being the start state and **D** being the goal state. Shown on the graph are path costs between states. The table lists the estimated distance from a state to the goal. Assume these estimates are admissible. Perform an **A* search** for this problem, showing **(i)** the tree that represents the nodes expanded and **(ii)** the **ordered** list of nodes to expand at each step **along with the node's *f* value.** (That list starts with {A90}, showing A's *f*-value of 90.) Recall that $f = g + h$. (5) [5]



Node	Estimated distance to goal
A	90
B	45
C	40
D	0

3. a) What do you mean by Adversarial Search. Solve 8-Queen problem using Genetic Algorithm or Hilll-climbing search. [5]

   b) Explain Alpha-Beta Pruning Idea. Apply this idea for the following example. [3]

   c) Illustrates the Wumpus World problem along with its PEAS. [4]

4. a) Elaborate the following terms with example. [4]
      i) Horn clause
      ii) Modus ponens
      iii) Disjunctive normal form (DNF)
      iv) Conjunctive normal form (CNF)

AVAILABLE AT:
Onebyzero Edu - Organized Learning, Smooth Career
The Comprehensive Academic Study Platform for University Students in Bangladesh (www.onebyzeroedu.com)